Chapter 2. Multivariate Random Variables

Joint and marginal distributions, moments, conditional expectation, independence, random vectors

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2.1 Joint and marginal distributions

A pair (X, Y) is described by a **joint pmf** $f(x, y) = P(X = x, Y = y)$ in the discrete case, or a **joint pdf** $f \geq 0$ with $P(a \leq X \leq b, c \leq Y \leq d) = \int_a^b \int_c^d f(x, y)\, dy\, dx$ in the continuous case. In both cases $f \geq 0$ and f sums or integrates to 1.

Definition (Marginal distributions)

$f_1(x) = \sum_y f(x, y)$ or $\int f(x, y)\, dy$, and $f_2(y) = \sum_x f(x, y)$ or $\int f(x, y)\, dx$.

Marginals are obtained by "summing out" the other variable; in a table they are the row and column totals. The joint distribution determines the marginals, but not conversely.

The **joint cdf** is $F(x, y) = P(X \leq x, Y \leq y)$. For continuous (X, Y), $f = \partial^2 F / \partial x \partial y$ at continuity points. The marginal cdfs are limits: $F_1(x) = \lim_{y \to \infty} F(x, y)$.

Theorem 2.1

(a) F is nondecreasing in each argument, and for $x_1 < x_2$, $y_1 < y_2$,

$$P(x_1 < X \leq x_2,\ y_1 < Y \leq y_2) = F(x_2, y_2) - F(x_1, y_2) - F(x_2, y_1) + F(x_1, y_1) \geq 0.$$

(b) $F(x, y) \to 0$ as $x \to -\infty$ or $y \to -\infty$, and $F \to 1$ as both $\to +\infty$. (c) F is right-continuous in each argument.

Proof. The rectangle identity is inclusion–exclusion for $\{X \leq x_2, Y \leq y_2\}$ minus the two strips. (b), (c) follow from continuity of probability (Theorem 1.2) as in one dimension. ∎

Example 2.1

Let $f(x, y) = c(x + y)$ on $0 \leq x, y \leq 1$. Then $\int_0^1 \int_0^1 (x + y)\, dy\, dx = 1$, so $c = 1$. The marginal is $f_1(x) = \int_0^1 (x + y)\, dy = x + \frac{1}{2}$ on $[0, 1]$, and by symmetry $f_2(y) = y + \frac{1}{2}$. Also $P(X + Y \leq 1) = \int_0^1 \int_0^{1-x} (x + y)\, dy\, dx = \int_0^1 \frac{1 - x^2}{2}\, dx = \frac{1}{3}$.

2.2 Expectations, covariance and correlation

$E[g(X, Y)] = \sum\sum g(x, y) f(x, y)$ or $\iint g f$; it is linear and monotone. If g depends only on x, $Eg(X)$ can be computed from the joint or from the marginal: integrating out y first gives $\int g(x) f_1(x)\, dx$.

Definition (Covariance and correlation)

$\operatorname{Cov}(X, Y) = E[(X - \mu_1)(Y - \mu_2)]$ and, for positive variances, $\rho = \operatorname{Corr}(X, Y) = \operatorname{Cov}(X, Y)/(\sigma_1 \sigma_2)$.

Theorem 2.2

(a) $\operatorname{Cov}(X, Y) = \operatorname{Cov}(Y, X)$ and $\operatorname{Cov}(X, X) = \operatorname{Var} X$. (b) $\operatorname{Cov}(aX + b, cY + d) = ac \operatorname{Cov}(X, Y)$. (c) $\operatorname{Cov}(X, Y) = E(XY) - E(X)E(Y)$.

Proof. (a) is immediate. (b) $aX + b - E(aX + b) = a(X - \mu_1)$, and similarly for Y. (c) Expand $(X - \mu_1)(Y - \mu_2)$ and use linearity. ∎

Theorem 2.3

(a) $\operatorname{Var}\left(\frac{Y-\mu_2}{\sigma_2} - \rho\frac{X-\mu_1}{\sigma_1}\right) = 1-\rho^2$. (b) $-1 \le \rho \le 1$. (c) $\rho = \pm 1$ iff $\frac{Y-\mu_2}{\sigma_2} = \pm\frac{X-\mu_1}{\sigma_1}$ with probability 1.

Proof. (a) Expand: $1 - 2\rho\cdot\rho + \rho^2 = 1-\rho^2$. (b) A variance is nonnegative. (c) $\rho = \pm 1$ makes the variance in (a) zero, so the variable is constant (zero, since its mean is zero) with probability 1 (Theorem 1.10). ■

Equivalently,

$$|\operatorname{Cov}(X,Y)| \le \sigma_1\sigma_2,$$

a case of the **Cauchy–Schwarz inequality**, $|E(VW)| \le \sqrt{EV^2\,EW^2}$. The correlation measures *linear* association, free of units.

Example 2.2 (Example 2.1 continued)

$EX = \int_0^1 x(x+\frac{1}{2})\,dx = \frac{7}{12}$, $EX^2 = \int_0^1 x^2(x+\frac{1}{2})\,dx = \frac{5}{12}$, so $\operatorname{Var} X = \frac{5}{12} - \frac{49}{144} = \frac{11}{144}$. Also $E(XY) = \int_0^1\int_0^1 xy(x+y)\,dy\,dx = \frac{1}{3}$, so $\operatorname{Cov}(X,Y) = \frac{1}{3} - \frac{49}{144} = -\frac{1}{144}$ and $\rho = -\frac{1}{11}$.

Joint moments and generating functions. The (r,s) joint moment is $E(X^rY^s)$. If $M(t_1,t_2) = E(e^{t_1X+t_2Y})$ is finite near $(0,0)$, then (a) all joint moments exist and $E(X^rY^s) = \partial^{r+s}M/\partial t_1^r\partial t_2^s$ at 0, and (b) M determines the joint distribution. The marginal mgfs are $M_X(s) = M(s,0)$ and $M_Y(t) = M(0,t)$. The joint cumulant generating function $C = \log M$ has $\partial C/\partial t_i(0)$ equal to the means and $\partial^2 C/\partial t_i\partial t_j(0)$ equal to the variances and covariance.

2.3 Conditional distributions and conditional expectation

Definition (Conditional pmf/pdf)

For $f_1(x) > 0$, $f_{2|1}(y \mid x) = f(x,y)/f_1(x)$. For each fixed x it is a pmf/pdf in y.

In the continuous case $P(X=x) = 0$, so this is justified as a limit: if f and f_1 are continuous,

$$P(c \le Y \le d \mid x \le X \le x+h) = \frac{\frac{1}{h}\int_x^{x+h}\int_c^d f(s,y)\,dy\,ds}{\frac{1}{h}\int_x^{x+h} f_1(s)\,ds} \xrightarrow{h\downarrow 0} \int_c^d f_{2|1}(y \mid x)\,dy.$$

Theorem 2.4

$P(a \le X \le b, c \le Y \le d) = \int_a^b P(c \le Y \le d \mid X = x) f_1(x)\,dx$ (a sum in the discrete case).

Proof. Substitute $f(x,y) = f_{2|1}(y \mid x) f_1(x)$ into the double integral and integrate in y first. ■

Definition (Conditional expectation and variance)

$E[g(X,Y) \mid X = x] = \int g(x,y) f_{2|1}(y \mid x)\,dy$; $\mu_{2|1}(x) = E(Y \mid X = x)$; $\operatorname{Var}(Y \mid X = x) = E[(Y - \mu_{2|1}(x))^2 \mid X = x] = E(Y^2 \mid X = x) - \mu_{2|1}(x)^2$. Replacing x by X gives the random variables $E(Y \mid X)$ and $\operatorname{Var}(Y \mid X)$.

Conditional expectation is linear and monotone, and functions of the conditioning variable factor out: $E[c(X)g(Y) \mid X = x] = c(x)E[g(Y) \mid X = x]$.

Theorem 2.5

(a) (Tower property) $E[E(Y \mid X)] = E(Y)$. (b) $\operatorname{Cov}(Y - E(Y \mid X), v(X)) = 0$ for every function v with $Ev(X)^2 < \infty$.

Proof. (a) $\int \left(\int y f_{2|1}(y \mid x)\, dy \right) f_1(x)\, dx = \iint y f(x,y)\, dy\, dx = EY$. (b) Let $Z = Y - E(Y \mid X)$. Then $E(Z \mid X) = 0$, so $EZ = 0$ by (a), and $E[Zv(X)] = E[v(X)E(Z \mid X)] = 0$. ■

Theorem 2.6 (Best predictor)

Among all functions u, $E[(Y - u(X))^2]$ is minimized by $u(X) = E(Y \mid X)$, and the minimum is $E[\operatorname{Var}(Y \mid X)]$.

Proof. Write $Y - u(X) = (Y - E(Y \mid X)) + v(X)$ with $v = E(Y \mid X) - u(X)$. The cross term has expectation 0 by Theorem 2.5(b), so $E(Y-u)^2 = E(Y - E(Y \mid X))^2 + Ev(X)^2$, minimized when $v = 0$. And $E(Y - E(Y \mid X))^2 = E[E((Y - E(Y \mid X))^2 \mid X)] = E \operatorname{Var}(Y \mid X)$. ■

$E(Y \mid X)$ is called the **regression function** of Y on X.

Theorem 2.7 (Variance decomposition)

$\operatorname{Var} Y = E[\operatorname{Var}(Y \mid X)] + \operatorname{Var}[E(Y \mid X)]$.

Proof. $Y - \mu = (Y - E(Y \mid X)) + (E(Y \mid X) - \mu)$; the two parts are uncorrelated by Theorem 2.5(b). Square and take expectations, using $E[E(Y \mid X)] = \mu$ and Theorem 2.6. ■

Geometrically $Y - EY$ splits into orthogonal pieces: the part explained by X and the residual. This is the basis of regression and ANOVA (Chapter 10).

Example 2.3 (Example 2.1 continued)

$f_{2|1}(y \mid x) = \frac{x+y}{x+1/2}$ on $[0,1]$, so $E(Y \mid X = x) = \frac{x/2+1/3}{x+1/2} = \frac{3x+2}{6x+3}$. The regression is not linear, and it decreases in x, consistent with $\rho < 0$.

2.4 Independence

Definition (Independent random variables)

X and Y are independent if $P(X \in A, Y \in B) = P(X \in A)P(Y \in B)$ for all (Borel) sets A, B.

Theorem 2.8

Each of the following is equivalent to independence: (a) $F(x,y) = F_1(x)F_2(y)$ for all x, y; (b) $f(x,y) = f_1(x)f_2(y)$ for all x, y; (c) $M(t_1,t_2) = M_X(t_1)M_Y(t_2)$ near 0 (when the mgf exists). Moreover (b) holds iff $f(x,y) = g_1(x)g_2(y)$ for *some* functions g_1, g_2.

Proof. Rectangles determine the distribution, so (a)$\Leftrightarrow$ independence. (a) and (b) are equivalent by differentiation and integration. (c) follows from the uniqueness of the mgf. If $f = g_1 g_2$, integrating gives $f_1 = g_1 \int g_2$, $f_2 = g_2 \int g_1$ and $1 = \int g_1 \int g_2$, so $f = f_1 f_2$. ■

The factorization must hold on the whole plane, *support included*: $f(x,y) = 8xy$ on $0 \le x \le y \le 1$ looks like a product, but the triangular support makes X, Y dependent.

Theorem 2.9

If X, Y are independent then (a) $g_1(X)$ and $g_2(Y)$ are independent; (b) $E[g_1(X)g_2(Y)] = Eg_1(X)\,Eg_2(Y)$; (c) $\operatorname{Cov}(X,Y) = 0$; (d) $\operatorname{Var}(X+Y) = \operatorname{Var} X + \operatorname{Var} Y$. In general $\operatorname{Var}(X+Y) = \operatorname{Var} X + \operatorname{Var} Y + 2\operatorname{Cov}(X,Y)$.

Proof. (a) $\{g_1(X) \in A\} = \{X \in g_1^{-1}(A)\}$. (b) Factor the double integral. (c) is (b) with identities and Theorem 2.2(c). (d) Expand the square. ■

Uncorrelated does not imply independent: if X is uniform on $\{-1, 0, 1\}$ and $Y = X^2$, then $\operatorname{Cov}(X,Y) = EX^3 = 0$, yet Y is a function of X.

2.5 Random vectors

For $\mathbf{X} = (X_1, \dots, X_k)'$ the joint pmf/pdf, marginals (sum or integrate out the other coordinates), cdf and expectations are defined as above. The **mean vector** is $E\mathbf{X} = (EX_1, \dots, EX_k)'$, and the **covariance matrix** is

$$\operatorname{Var}(\mathbf{X}) = E[(\mathbf{X}-\mu)(\mathbf{X}-\mu)'] = \big(\operatorname{Cov}(X_i, X_j)\big)_{i,j}.$$

More generally $\operatorname{Cov}(\mathbf{X}, \mathbf{Y}) = E[(\mathbf{X}-\mu)(\mathbf{Y}-\eta)']$.

Theorem 2.10

For constant matrices A, C and vectors $\mathbf{b}, \mathbf{d}$:

(a) $E(A\mathbf{X}+\mathbf{b}) = AE\mathbf{X} + \mathbf{b}$;

(b) $\operatorname{Var}(A\mathbf{X}+\mathbf{b}) = A\operatorname{Var}(\mathbf{X})A'$ and $\operatorname{Cov}(A\mathbf{X}+\mathbf{b}, C\mathbf{Y}+\mathbf{d}) = A\operatorname{Cov}(\mathbf{X},\mathbf{Y})C'$;

(c) $\operatorname{Var}(\mathbf{X}+\mathbf{Y}) = \operatorname{Var}\mathbf{X} + \operatorname{Var}\mathbf{Y} + \operatorname{Cov}(\mathbf{X},\mathbf{Y}) + \operatorname{Cov}(\mathbf{Y},\mathbf{X})$;

(d) $\operatorname{Var}(\mathbf{X})$ is symmetric and nonnegative definite.

Proof. (a)–(c) follow from $E(CWD) = CE(W)D$ for a random matrix W, applied to $(\mathbf{X}-\mu)(\mathbf{Y}-\eta)'$. (d) $\mathbf{a}'\operatorname{Var}(\mathbf{X})\mathbf{a} = \operatorname{Var}(\mathbf{a}'\mathbf{X}) \geq 0$. ■

By the spectral theorem $\operatorname{Var}\mathbf{X} = P\operatorname{diag}(\lambda_1, \dots, \lambda_k)P'$ with P orthogonal and $\lambda_1 \geq \cdots \geq \lambda_k \geq 0$. Then $P'\mathbf{X}$ has uncorrelated coordinates with variances λ_i: these are the **principal components**, and the eigenvectors give the main directions of spread.

Generating functions. $M(\mathbf{t}) = E(e^{\mathbf{t}'\mathbf{X}})$ yields all joint moments and determines the distribution. The cgf $C = \log M$ gives $E\mathbf{X} = \nabla C(\mathbf{0})$ and $\operatorname{Var}\mathbf{X} = \nabla^2 C(\mathbf{0})$ (gradient and Hessian).

Conditioning. $f_{\mathbf{Y}|\mathbf{X}}(\mathbf{y} \mid \mathbf{x}) = f_{\mathbf{X},\mathbf{Y}}(\mathbf{x},\mathbf{y})/f_{\mathbf{X}}(\mathbf{x})$. The tower property, the orthogonality $\operatorname{Cov}(\mathbf{Y} - E(\mathbf{Y} \mid \mathbf{X}), v(\mathbf{X})) = 0$, the matrix decomposition

$$\operatorname{Var}\mathbf{Y} = E[\operatorname{Var}(\mathbf{Y} \mid \mathbf{X})] + \operatorname{Var}[E(\mathbf{Y} \mid \mathbf{X})],$$

and the best-predictor property for $E\|\mathbf{Y} - u(\mathbf{X})\|^2$ carry over with the same proofs.

Independence. $X_1, \dots, X_n$ are independent iff the joint pdf (or the mgf) factors into marginals, again iff it factors into *some* functions of the separate arguments. Then functions of separate components are independent, expectations of products factor, the covariance matrix of $\sum_i \mathbf{X}_i$ is $\sum_i \operatorname{Var}\mathbf{X}_i$, and $M_{\sum \mathbf{X}_i}(\mathbf{t}) = \prod_i M_{\mathbf{X}_i}(\mathbf{t})$.

Example 2.4

Let X_1, X_2, X_3 be i.i.d. standard exponential and $Y_1 = X_1$, $Y_2 = X_1 + X_2$. Then $\mathrm{Var}(Y_1, Y_2)' = A\, I\, A'$ with $A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$, i.e. $\begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$, so $\mathrm{Corr}(Y_1, Y_2) = 1/\sqrt{2}$.